

A Packing of 5 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 5 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.999999. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.879908, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 5 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.999999 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 5$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.879908, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 5/4 = 3.000000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N5.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 60 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
149999999758625841548484427638254404121496269388488732429560022425347570
331034225836732715101851736700637638868320727025121716483885579026654174
639345404258446842757696337013681161295273047461
```

denominator:

```
500000000001044752078643246667182341076770492574426193004137217142759744
632626820532238275217271929029119422734901365709405225022631086916998409
68184635517618008454405266361575742452687477692
```

B Coordinates

5 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N5.txt`.

```
-96534669436 -377455532869 22409374330 -81342372171-249740265931 15064863344 198994737202 11315493221722259787652 -114414161351 378485620201 -455845080682
293910427671 72547840937 -31146722862 -329962120737-316255250506 172699847436 554310737483 -171851001565699059611 329219093452 -19685599788 -124751251725
-543348430780 -262910897567 183711580327 -44325118978064906460 484821279103 80540471737 561027494446 -191022317906 17154440270 261987308830 -240409129733
-203186445953 -309674918854 -236272486973 -408194423307070970646 -428048336368 -230624084975 -3442462623256460462 172169050874 203419656376 -390728451419
-254521028345 -291452310788 420152736695 -40360260302167680049 420500471296 -116430281878 -1845738239061367749889 680732009565 319538253815 259927173213
-272379106375 -8252809602 -299610947676 48362639098707697384598 474399378694 -158077407692 -130134377919971471 -318721440979 -342717227962 -280716363129
237581366383 -613157990391 -170493612016 430476305561608306668 42051830856 239240907592 544833654153 -4996380388 -686370013739 155828431218 6536034257
-443016153940 409047897708 -397262776001 273596082337637081464 -479283764215 -155382821736 348579120034299535025992 -55803603390 -526122628407 457312639382
265049970810 167004200985 445367693454 12664444626388498891989 403325553538 37768023260 215938403288 379700715616 -129994582288 137594213776 412115880966
-325352781243 176231582809 507365607155 498748356339468350542893 -35759332169 -489376966933 -1735549984300140949589 -53981454506 133250744483 466010148033
-279396311481 -345385393804 301311464015 -667296848335009204097 -121912690942 592246059404 -488656576738625094868 257045102980 -537039979047 232655049817
158028250482 -179161824348 530410572170 39292649438341364546131 -34997155956 -180907936560 28938535016040402989502 527447701967 202500541287 -194393248856
485705499157 131317779771 231182009463 -1796443515251809974514 58139355329 -449987450022 -113406708855965800144 -74581537227 -321867073766 -360038745440
-384409155921 -403412539432 -188936831706 52872696313817147551 202527586311 -493264560106 22495821103318236997290 -326088093157 -435477229580 -331407723595
-454017637981 126225801093 -14195796818 -301270935288505272247 -323461589268 458535605927 21158930281803921301227 -212670824851 178703949309 7744093083
133684947111 -230829098441 420250632825 -3783312982036050110780 370289417895 -167638763481 -4694098890404536741408 -67613460356 -87362525495 -269037946845
337801131524 -578191186855 -92604519120 62764283931292933994918 29141543376 -212407223747 -322632034353
```